Code No: 118BJ **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July/August - 2022
EMBEDDED SYSTEMS
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
- - -

1.a) Draw and explain the typical Embedded system architecture.
b) Illustrate an application-specific embedded system with suitable example. [8+7]

2.a) Discuss about 8051 family series microcontrollers in expanding the memory.
b) What are the interrupts of 8051? Explain them briefly. [8+7]

3. Explain the different assembly programming tools used in 8051 microcontroller in detail. [15]

4.a) Write 8051 program to generate 2 KHz square wave on pin P1.0 using Timer interrupt.
b) Explain the concept of addressing modes in 8051 microcontroller. [8+7]

5.a) Describe the mail boxes and message queues with examples.
b) Explain about interrupt routines in an RTOS Environment. [8+7]

6.a) Discuss about the serial data communication.
b) What is a semaphore? Explain semaphores with example. [8+7]

7.a) List and describe the debugging tools used in an embedded system.
b) Briefly discuss about testing on host machine. [8+7]

8.a) Explain the concept of instruction level parallelism.
b) Explain the feature of ARM controller in detail. [8+7]

---oo0oo---

JNTUHUbeseExamPapers2022